QPCODE: **Reg.No : .....**

Name : .....

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) EXAMINATION OCTOBER 2026

FIFTH SEMESTER

MG5DSCMBG301 – ESSENTIALS OF IMMUNOLOGY

(2024 ADMISSION ONWARDS)

Duration: 1.5 Hours **Maximum Marks: 50**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

Short Notes Questions

Answer any **10** questions

Each question carries **2** marks

1.	What is variolation?	[K] / [CO:1]
2.	Define a hapten.	[K] / [CO:2]
3.	Mention any two differences between primary and secondary immune responses	[K] / [CO:3]
4.	What is cross-infection? Mention one example.	[U] / [CO:1]
5.	Which immunoglobulin is mainly present in secretions such as saliva and tears?	[K] / [CO:2]
6.	Mention any two functions of antibodies in humoral immunity.	[U] / [CO:3]
7.	What is the role of skin in innate immunity?	[K] / [CO:1]
8.	What type of antigen participates in an agglutination reaction?	[U] / [CO:2]
9.	Give two examples of organ-specific autoimmune diseases.	[K] / [CO:3]
10.	Give two examples of artificially acquired passive immunity.	[K] / [CO:1]
11.	What are antigen-presenting cells (APCs)? Give two examples.	[U] / [CO:1]

12. What does the absence of haemolysis indicate in direct CFT? [U] / [CO:2]

[2x10 = 20]

Part B

Short Essay Questions

Answer any **4** questions

Each question carries **5** marks

13. Explain the mechanism of Type I hypersensitivity. [U] / [CO:3]

14. Explain the structure and functions of the thymus. [U] / [CO:1]

15. Describe the factors responsible for the development of autoimmune diseases. [U] / [CO:3]

16. Describe the steps involved in the production of monoclonal antibodies. [U] / [CO:2]

17. Explain Type IV hypersensitivity and why it is called delayed hypersensitivity. [U] / [CO:3]

18. Explain the role of $CD4^+$ and $CD8^+$ T cells in cell-mediated immunity. [An] / [CO:3]

[5x4 = 20]

Part C

Essay Question

Answer any **1** questions

Each question carries **10** marks

19. Explain the classical pathway of complement activation in detail with a diagram. [U] / [CO:2]

20. Describe the mechanism of Type III hypersensitivity and explain the role of immune complexes in tissue damage. [U] / [CO:3]

[10x1 = 10]